Exponentials & Logarithms

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Logarithm definition and evaluation**

Rewrite $\log_3 81 = 4$ in exponential form, and verify.

$3^4 = 81$, which checks out.

2 **[Foundational]** • **Target: Logarithm laws**

Simplify $\log 8 + \log 5$ as a single logarithm.

$\log(8 \times 5) = \log 40$.

3 **[Foundational]** • **Target: Logarithm laws**

Write $3\log_2 x + \log_2 y$ as a single logarithm.

$\log_2 (x^3 y)$.

4 **[Foundational]** • **Target: Exponential growth and decay models**

A population of bacteria starts at 300 and doubles every 4 hours. Write an exponential model and find the population after 12 hours.

$P(t) = 300 \cdot 2^{t/4}$. $P(12) = 300 \cdot 2^3 = 300 \times 8 = 2400$.

5 **[Foundational]** • **Target: Exponential growth and decay models**

A car worth 24{,}000 dollars depreciates at 12 percent per year. Find its value after 3 years, to the nearest dollar.

$V = 24000(0.88)^3 = 24000 \times 0.681472 \approx 16{,}355$ dollars.

6 **[Foundational]** • **Target: Graphs of exponential functions**

The graph of $y = 2^x$ is shown below. Use the graph to estimate the value of x for which $y = 8$, then confirm your answer algebraically.

From the graph, $x \approx 3$. Algebraically: $2^x = 8 = 2^3$, so $x = 3$.

7 **[Foundational]** • **Target: Natural logarithms and e**

Solve $e^x = 20$ for x, to the nearest hundredth, using natural logarithms.

$x = \ln(20) \approx 2.996 \approx 3.00$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: A well-designed word problem: comparing two investment accounts**

This question has three parts. An investment of 5000 dollars is placed in an account that grows according to $A(t) = 5000e^{0.04t}$, where t is measured in years. (a) Find the value of the investment after 10 years, to the nearest dollar. (b) Find how many years it takes for the investment to double, to the nearest tenth of a year, using natural logarithms. (c) A second account offers simple (non-compounded) growth of 5 percent per year on the same initial 5000 dollars. Determine which account has the greater value after 10 years, and by how much, to the nearest dollar.

(a) $A(10) = 5000e^{0.4} \approx 5000(1.4918) \approx 7459$ dollars. (b) Set $10000 = 5000e^{0.04t}$, so $2 = e^{0.04t}$, giving $\ln 2 = 0.04t$, so $t = \frac{\ln 2}{0.04} \approx \frac{0.6931}{0.04} \approx 17.3$ years. (c) Simple growth value: $5000 + 5000(0.05)(10) = 5000 + 2500 = 7500$ dollars. Since $7500 > 7459$, the simple-growth account is greater after 10 years, by about $7500 - 7459 = 41$ dollars.

9 **[Intermediate]** • **Target: A well-designed word problem: comparing two investment accounts**

Solve the equation $5e^{0.04t} = 45$ for t in terms of natural logarithms, and find its value rounded to two decimal places.

Step 1: Isolate the exponential expression: $e^{0.04t} = \frac{45}{5} = 9$. Step 2: Take the natural logarithm of both sides: $\ln(e^{0.04t}) = \ln(9) \Rightarrow 0.04t = \ln(9)$. Step 3: Solve for t: $t = \frac{\ln(9)}{0.04} = 25\ln(9) = 25\ln(3^2) = 50\ln(3) \approx 50(1.0986) = \mathbf{54.93}$. The exact solution is $\mathbf{50\ln(3)}$ (approximately 54.93).

10 **[Intermediate]** • **Target: Exponent rules**

Solve the logarithmic equation $\log_3(x^2 - 7) = 2$ for all real solutions, and verify that all roots lie within the valid domain.

Step 1: Convert to exponential form: $x^2 - 7 = 3^2 = 9$. Step 2: Solve the quadratic equation: $x^2 = 9 + 7 = 16 \Rightarrow x = \pm 4$. Step 3: Verify the domain $x^2 - 7 > 0$: For both $x = 4$ and $x = -4$: $x^2 - 7 = 16 - 7 = 9 > 0$. Both roots are valid. The solutions are $\mathbf{x = 4}$ and $\mathbf{x = -4}$.

11 **[Intermediate]** • **Target: Solving exponential equations**

Solve the exponential equation $3^{2x-1} = 5^{x+2}$ for x in terms of natural logarithms.

Step 1: Take the natural logarithm of both sides: $\ln(3^{2x-1}) = \ln(5^{x+2}) \Rightarrow (2x - 1)\ln(3) = (x + 2)\ln(5)$. Step 2: Expand and group all terms containing x: $2x\ln(3) - \ln(3) = x\ln(5) + 2\ln(5) \Rightarrow x[2\ln(3) - \ln(5)] = 2\ln(5) + \ln(3)$. Step 3: Solve for x: $x = \frac{\mathbf{2\ln(5) + \ln(3)}}{\mathbf{2\ln(3) - \ln(5)}} = \frac{\ln\mathbf{(75)}}{\ln\mathbf{(9/5)}} \approx 7.37$.

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12 [Intermediate] • Target: Logarithm definition and evaluation

Simplify the expression $\left(\frac{16x^8y^{-4}}{81z^{12}}\right)^{-3/4}$ and write your answer using only positive exponents.

Step 1: Invert the fraction to remove the negative exponent: $\left(\frac{16x^8y^{-4}}{81z^{12}}\right)^{-3/4} = \left(\frac{81z^{12}}{16x^8y^{-4}}\right)^{3/4} = \left(\frac{81y^4z^{12}}{16x^8}\right)^{3/4}$. Step 2: Take the 4th root of each factor: $\left(\frac{81y^4z^{12}}{16x^8}\right)^{1/4} = \frac{81^{1/4}(y^4)^{1/4}(z^{12})^{1/4}}{16^{1/4}(x^8)^{1/4}} = \frac{3yz^3}{2x^2}$. Step 3: Cube the result.

13 [Intermediate] • Target: Logarithm laws

A population of bacteria in a petri dish starts with 250 organisms and quadruples every 5 hours. How many hours will it take for the population to reach 16,000?

Step 1: Set up the exponential growth equation: $P(t) = 250 \times 4^{t/5} = 16000$. Step 2: Divide both sides by 250: $4^{t/5} = \frac{16000}{250} = \frac{1600}{25} = 64$. Step 3: Express 64 as a power of 4: $64 = 4^3 \Rightarrow 4^{t/5} = 4^3 \Rightarrow \frac{t}{5} = 3 \Rightarrow t = 15$ hours. It will take exactly 15 hours.

14 [Intermediate] • Target: Solving logarithmic equations

Two assembly line components operate independently. Component A has a 95% reliability rate (probability of functioning properly is 0.95), and Component B has a 90% reliability rate (probability of functioning is 0.90). What is the probability that at least one of the two components fails?

Step 1: Use the complement rule. The event "at least one fails" is the exact complement of "both components function properly": $P(\geq 1 \text{ fails}) = 1 - P(\text{both function})$. Step 2: Since the components operate independently: $P(\text{both function}) = P(A) \times P(B) = 0.95 \times 0.90 = 0.855$. Step 3: Calculate the complement: $P(\geq 1 \text{ fails}) = 1 - 0.855 = \mathbf{0.145\ (14.5\%)}$. The probability that at least one component fails is 0.145 (or 14.5%).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Exponential growth and decay models • Cross-Lesson: 1.01 Heart of Algebra

An investment fund gains 30% in value during Year 1, decreases by 20% in Year 2, and increases by 10% in Year 3. Determine the overall net percent change of the fund across the three-year period.

Step 1: Express each yearly change as a decimal multiplier:Year 1 (+30%): $1 + 0.30 = 1.30$Year 2 (-20%): $1 - 0.20 = 0.80$Year 3 (+10%): $1 + 0.10 = 1.10$ Step 2: Multiply the factors sequentially: $M_{\text{net}} = 1.30 \times 0.80 \times 1.10 = 1.04 \times 1.10 = 1.144$. Step 3: Convert to net percentage change: $\text{Net Percent Change} = (1.144 - 1) \times 100\% = \mathbf{+14.4\%}$. The investment gained an overall 14.4%.

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16 [Challenging] • Target: Graphs of exponential functions • Cross-Lesson: 2.01 Quadratics & Polynomials

Two angles of a triangle are 55° and 75°. Find the third.

Step 1: Use triangle angle sum theorem: The three interior angles of any triangle sum to 180°: $\theta + 55^\circ + 75^\circ = 180^\circ$ Step 2: Solve for θ: $\theta + 130^\circ = 180^\circ \Rightarrow \theta = 180^\circ - 130^\circ = \mathbf{50}^\circ$ The third angle measures $\mathbf{50}^\circ$.

17 [Challenging] • Target: Natural logarithms and e • Cross-Lesson: 3.01 Functions & Graphs

A solid metal cone with base radius 4 cm and height 9 cm is submerged into a cylindrical beaker with internal radius 6 cm partially filled with water. Assuming the cone is fully submerged and no water overflows, by how many centimeters does the water level in the beaker rise?

Step 1: Compute the volume of the submerged cone: $V_{\text{cone}} = \frac{1}{3}\pi r_{\text{cone}}^2 h_{\text{cone}} = \frac{1}{3}\pi(4^2)(9) = \frac{1}{3}\pi(16)(9) = 48\pi$ cm^3. Step 2: The volume of water displaced forms a cylinder with radius $R = 6$ cm and height Δh: $V_{\text{displaced}} = \pi R^2(\Delta h) = \pi(6^2)\Delta h = 36\pi\Delta h$. Step 3: Equate volumes and solve for Δh.

18 [Challenging] • Target: A well-designed word problem: comparing two investment accounts • Cross-Lesson: 4.01 Ratios, Rates & Data

Consider the ordered dataset of 12 numbers: 14, 16, 18, 20, 22, 24, 26, 28, 30, 34, 38, 72. (a) Determine Q_1, the median, Q_3, and the IQR. (b) Use Tukey's $1.5 \times$ IQR rule to determine whether 72 is an outlier.

(a) Five-Number Summary Calculations:Lower half (first 6 values): 14, 16, 18, 20, 22, 24 $\Rightarrow Q_1 = \frac{18+20}{2} = \mathbf{19}$.Median of all 12 values: $\frac{24+26}{2} = \mathbf{25}$.Upper half (last 6 values): 26, 28, 30, 34, 38, 72 $\Rightarrow Q_3 = \frac{30+34}{2} = \mathbf{32}$.IQR $= Q_3 - Q_1 = 32 - 19 = \mathbf{13}$. (b) Outlier Boundary Check:

Upper Outlier Boundary $= Q_3 + 1.5 \times \text{IQR} = 32 + 1.5(13) = 32 + 19.5 = \mathbf{51.5}$. Since $72 > 51.5$, the va.

19 [Challenging] • Target: Exponent rules • Cross-Lesson: 5.01 Statistics & Probability

An equilateral triangle has a side length of 12 cm. Find: (a) the exact length of its altitude, and (b) its exact area.

(a) Altitude using $30^\circ - 60^\circ - 90^\circ$ triangle: Dropping an altitude splits the equilateral triangle into two congruent $30^\circ - 60^\circ - 90^\circ$ right triangles with short leg $b/2 = 6$ cm and hypotenuse 12 cm. The altitude is opposite the 60° angle: $h = 6\sqrt{3}$ cm. (b) Area formula: $A = \frac{1}{2}bh = \frac{1}{2}(12)(6\sqrt{3}) = \mathbf{36\sqrt{3}}$ cm^2. (Alternatively.

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20 [Challenging] • Target: Solving exponential equations • Cross-Lesson: 6.01 Geometry

A rectangular water storage reservoir has dimensions 2.5 meters by 3 meters by 1.2 meters. A drainage pump empties the reservoir at a constant rate of 75 liters per minute. Given that $1\ \text{m}^3 = 1000\ \text{liters}$, how many hours will it take to completely empty the full reservoir?

Step 1: Find the reservoir volume in cubic meters: $V = 2.5 \times 3.0 \times 1.2 = 9.0\ \text{m}^3$. Step 2: Convert volume to liters: $V = 9.0\ \text{m}^3 \times 1000\ \text{L/m}^3 = 9000\ \text{liters}$. Step 3: Compute drainage time in minutes: $t_{\text{min}} = \frac{9000\ \text{L}}{75\ \text{L/min}} = 120\ \text{minutes}$. Step 4: Convert to hours: $t = \frac{120\ \text{min}}{60\ \text{min/h}} = \mathbf{2}\ \text{hours}$. It takes exactly 2 hours.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Exponent rules

A population of bacteria starts at 488 and doubles every 6 hours. Find the population after 12 hours.

a. 2440

b. 1952 ✓

c. 976

d. 2928

$P(t) = 488 \cdot 2^{t/6}$. $P(12) = 488 \cdot 2^2 = 488 \times 4 = 1952$.

22 [Foundational] • Target: Solving exponential equations

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 6)$ and $(2, 18)$. Determine the value of a.

a. **2** ✓

b. 3

c. 6

d. 9

Let the function be $f(x) = a \cdot b^x$. We are given two points that lie on the graph: 1. $(1, 6)$: Substituting these values into the function equation, we get $6 = a \cdot b^1$, which simplifies to $6 = ab$. 2. $(2, 18)$: Substituting these values into the function equation, we get $18 = a \cdot b^2$. We now have a system of two equations with two unknowns, a and b: (1) $ab = 6$ (2) $ab^2 = 18$ To solve for a and b, we can divide equation (2) by equation (1): $\frac{ab^2}{ab} = \frac{18}{6}$ $b = 3$ Now.

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23 [Foundational] • Target: Logarithm definition and evaluation

The graph of the exponential function $f(x) = a^x$ passes through the point $(2, 9)$. What is the value of a?

a. 3 ✓

b. 9

c. $\sqrt{3}$

d. 81

The function is given by $f(x) = a^x$. We are told that the graph passes through the point $(2, 9)$. This means that when $x = 2$, $f(x) = 9$. Substituting these values into the function definition, we get: $9 = a^2$ To find the value of a, we take the square root of both sides: $a = \sqrt{9}$ Since a represents the base of an exponential function, it must be positive. Therefore, $a = 3$. Common misconceptions might arise from not understanding what it means for a graph to pass through a point or from incorrec.

24 [Foundational] • Target: Logarithm laws

The population of a species is initially 2500 and increases by 8% each year. Which function models the population $P(t)$ after t years?

a. $P(t) = 2500 + (1.08)^t$

b. $P(t) = 2500(-7)^t$

c. $P(t) = 2500(8)^t$

d. $P(t) = 2500(1.08)^t$ ✓

The standard exponential growth or decay formula is $P(t) = P_0(1 \pm r)^t$, where P_0 is the initial amount and r is the annual rate of change as a decimal. Here, initial population $P_0 = 2500$. Rate: $r = 8\% = 0.08$. Growth/decay factor: $1 + 0.08 = 1.08$. Thus, the population function is $P(t) = 2500(1.08)^t$.

25 [Foundational] • Target: Solving logarithmic equations

The population of a species is initially 4000 and decreases by 10% each year. Which function models the population $P(t)$ after t years?

a. $P(t) = 4000(11)^t$

b. $P(t) = 4000 + (0.9)^t$

c. $P(t) = 4000(10)^t$

d. $P(t) = 4000(0.9)^t$ ✓

The standard exponential growth or decay formula is $P(t) = P_0(1 \pm r)^t$, where P_0 is the initial amount and r is the annual rate of change as a decimal. Here, initial population $P_0 = 4000$. Rate: $r = 10\% = 0.1$. Growth/decay factor: $1 - 0.1 = 0.9$. Thus, the population function is $P(t) = 4000(0.9)^t$.

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26 [Foundational] • Target: Exponential growth and decay models

The graph of the function $f(x) = a \cdot b^x$ is shown below. If the graph passes through the points $(1, 12)$ and $(3, 108)$, what is the value of a?

a. 2

b. 3

c. **4** ✓

d. 6

Let the function be $f(x) = a \cdot b^x$. Given that the graph passes through $(1, 12)$ and $(3, 108)$, we can set up a system of equations: 1) For $(1, 12)$: $12 = a \cdot b^1 \Rightarrow 12 = ab$ 2) For $(3, 108)$: $108 = a \cdot b^3$ From equation (1), we can express a in terms of b: $a = \frac{12}{b}$. Substitute this expression for a into equation (2): $108 = \left(\frac{12}{b}\right) \cdot b^3$ $108 = 12b^2$ Divide both sides by 12: $b^2 = \frac{108}{12}$ $b^2 = 9$ Since b is the base of an exponen.

27 [Foundational] • Target: Graphs of exponential functions

Consider the function $f(x) = 3^x - 1$. What is the value of $f(2)$?

a. 5

b. 6

c. **8** ✓

d. 9

To find the value of $f(2)$, we substitute $x = 2$ into the function $f(x) = 3^x - 1$. $f(2) = 3^2 - 1$ $f(2) = 9 - 1$ $f(2) = 8$ Therefore, the value of $f(2)$ is 8.

28 [Foundational] • Target: Natural logarithms and e

A particle's position x in meters along a straight line is described by the equation $x(t) = e^{2t} - 6e^t + 5$, where t is time in seconds. The particle starts at $t = 0$. Find the time $t > 0$ when the particle momentarily stops, and the total distance traveled by the particle from $t = 0$ until it next comes to rest.

a. Time = $\ln 3$ s, Distance = 4 m

b. Time = $\ln 2$ s, Distance = 1 m

c. Time = $\ln 3$ s, Distance = 2 m

d. **Time =** $\ln 2$ **s, Distance =** 2 **m** ✓

The particle's position is given by $x(t) = e^{2t} - 6e^t + 5$. To find when the particle momentarily stops, we need to find the velocity $v(t)$, which is the derivative of the position with respect to time. $v(t) = \frac{dx}{dt} = 2e^{2t} - 6e^t$. The particle momentarily stops when $v(t) = 0$. So, $2e^{2t} - 6e^t = 0$. Let $u = e^t$. Then the equation becomes $2u^2 - 6u = 0$, which factors as $2u(u - 3) = 0$. This gives solutions $u = 0$ or $u = 3$. Since $u = e^t$, $e^t = 0$ has no solution, so we have $e^t = 3$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: A well-designed word problem: comparing two investment accounts

If $9^{x+1} = 27^{x-1}$, what is the value of x?

a. 2

b. 3

c. 5 ✓

d. 7

To solve the equation $9^{x+1} = 27^{x-1}$, express both sides in terms of a common base: 1. Recognize that both 9 and 27 are powers of 3: $9 = 3^2$ and $27 = 3^3$ 2. Substitute these into the original equation: $(3^2)^{x+1} = (3^3)^{x-1}$ 3. Use the power of a power rule, $(a^m)^n = a^{mn}$, distributing across both terms in each exponent: $3^{2(x+1)} = 3^{3(x-1)}$ $3^{2x+2} = 3^{3x-3}$ 4. Since the bases are equal and positive (and not equal to 1), equate the exponents: $2x + 2 = 3x - 3$.

30 [Intermediate] • Target: Exponent rules

Which of the following is equivalent to $\log_b\left(\frac{x^5y^3}{z^4}\right)$ for all positive values of x, y, z?

a. $5\log_b(x) \cdot 3\log_b(y) - 4\log_b(z)$

b. $\frac{5\log_b(x) + 3\log_b(y)}{4\log_b(z)}$

c. $5\log_b(x) - 3\log_b(y) - 4\log_b(z)$

d. $5\log_b(x) + 3\log_b(y) - 4\log_b(z)$ ✓

Apply the fundamental logarithmic laws: 1) Quotient Rule: $\log_b(M/N) = \log_b(M) - \log_b(N)$ $\log_b\left(\frac{x^5y^3}{z^4}\right) = \log_b(x^5y^3) - \log_b(z^4)$ 2) Product Rule: $\log_b(MN) = \log_b(M) + \log_b(N)$ $= \log_b(x^5) + \log_b(y^3) - \log_b(z^4)$ 3) Power Rule: $\log_b(M^k) = k\log_b(M)$ $= 5\log_b(x) + 3\log_b(y) - 4\log_b(z)$.

31 [Intermediate] • Target: Solving exponential equations

For all positive values of x, which expression is equivalent to $x^{\frac{4}{5}}$?

a. $\frac{x^4}{5}$

b. $\sqrt[5]{x^4}$ ✓

c. $\frac{1}{\sqrt[5]{x^4}}$

d. $\sqrt[4]{x^5}$

By the definition of rational exponents: $x^{\frac{m}{n}} = \sqrt[n]{x^m} = (\sqrt[n]{x})^m$ where n is the index of the radical (root) and m is the power. Thus, $x^{\frac{4}{5}} = \sqrt[5]{x^4}$.

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32 [Intermediate] • Target: Logarithm definition and evaluation

What is the exact value of $\log_3\left(\frac{1}{3}\right)$?

a. -1 ✓

b. 1

c. -3

d. 0

By definition, $\log_b(y) = x$ is equivalent to $b^x = y$. Here, we set $\log_3\left(\frac{1}{3}\right) = x$, which means: $3^x = \frac{1}{3} = 3^{-1}$ Since the bases are identical, $x = -1$.

33 [Intermediate] • Target: Logarithm laws

In 2011, the population of Cairo was 9.12 million. Assuming the population increased by 1% each year, what was the approximate population of Cairo in 2022?

a. 10.074154 million

b. 10.123200 million

c. **10.174896 million** ✓

d. 10.276645 million

The elapsed time is $2022 - 2011 = 11$ years. Using exponential growth: $P = 9.12(1 + 0.01)^{11} = 9.12(1.01)^{11} \approx 10.174896$ million.

34 [Intermediate] • Target: Solving logarithmic equations

The number of bees in a beehive rose from 2400 to 2520 in one month. How many bees are expected in the next month, assuming the population continues to experience the same monthly percentage increase?

a. 2600

b. 2640

c. **2646** ✓

d. 2656

Monthly growth factor: $\frac{2520}{2400} = 1.05$ (a 5% increase). For the following month: $2520 \times 1.05 = 2646$ bees.

35 [Intermediate] • Target: Exponential growth and decay models

A population starts at 140 and doubles every year. What is the population after 1 years?

a. 140

b. **280** ✓

c. 290

d. 420

Applying Exponentials & Logarithms principles, $A = 140 \times 2^1 = 280$ Therefore, the verified correct option is 280.

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36 [Intermediate] • Target: Graphs of exponential functions

A population starts at 50 and triples every year. What is the population after 1 years?

a. 200

b. 100

c. 160

d. **150** ✓

Applying Exponentials & Logarithms principles, $A = 50 \times 3^{1} = 150$ Therefore, the verified correct option is 150.

37 [Intermediate] • Target: Natural logarithms and e

Given the equation $\log_3(x^2) = 6$, what is the positive value of x?

a. 27 ✓

b. 9

c. 81

d. 3

Using logarithmic properties and converting to exponential form gives the solution 27.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: comparing two investment accounts • Cross-Lesson: 4.01 Ratios, Rates & Data

Given the equation $2\log_3(x) = \log_3(8x - 15)$, what is the product of all valid solutions for x?

a. 15 ✓

b. 8

c. 5

d. 3

Using logarithmic properties and converting to exponential form gives the solution 15.

39 [Challenging] • Target: Exponent rules • Cross-Lesson: 5.01 Statistics & Probability

Given the equation $\log_2(x) + \log_2(x - 3) = 2$, what is the x?

a. 4 ✓

b. 1

c. -1

d. 2

Using logarithmic properties and converting to exponential form gives the solution 4.

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40 [Challenging] • Target: Solving exponential equations • Cross-Lesson: 6.01 Geometry

Given the equation $3^{2x} - 10 \cdot 3^x + 9 = 0$, what is the sum of all real solutions for x?

a. 2 ✓

b. 0

c. 1

d. 3

Using logarithmic properties and converting to exponential form gives the solution 2.

41 [Challenging] • Target: Logarithm definition and evaluation • Cross-Lesson: 7.01 Trigonometry & Additional Topics

Given the equation $2^{2x} - 9 \cdot 2^x + 8 = 0$, what is the larger real solution for x?

a. 3 ✓

b. 0

c. 4

d. 2

Using logarithmic properties and converting to exponential form gives the solution 3.

42 [Challenging] • Target: Logarithm laws • Cross-Lesson: 1.01 Heart of Algebra

A radioactive sample of 1900 grams has a half-life of 1 year. How much remains after 2 years?

a. 480 g

b. 476 g

c. 950 g

d. **475 g** ✓

$A = 1900(\frac{1}{2})^2 = 475$ g

43 [Challenging] • Target: Solving logarithmic equations • Cross-Lesson: 2.01 Quadratics & Polynomials

Given $\log_{10}(7) \approx p$, what does $\log_{10}(7^4)$ equal, in terms of p?

a. p+4

b. **4p** ✓

c. p^4

d. p/4

$\log_{10}(7^4) = 4\log_{10}(7) = 4p$

44 [Challenging] • Target: Exponential growth and decay models • Cross-Lesson: 3.01 Functions & Graphs

A population of 500 grows at 10% per year. Which expression gives its size after t years?

a. $500 + 10t$

b. $500(1.1)^t$ ✓

c. $500(0.1)^t$

d. $500 \cdot 1.1t$

Exponential growth: $P = 500(1.1)^t$.

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45 [Challenging] • Target: Graphs of exponential functions • Cross-Lesson: 4.01 Ratios, Rates & Data

Two angles of a triangle are 48° and 73°. What is the third angle?

a. 49°

b. 59° ✓

c. 69°

d. 79°

Applying Geometry principles, $180 - 48 - 73 = 59^\circ$. Therefore, the verified correct option is 59°.

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